Pre-alpha build

Jeremiah Dados, Dmytro Stavskyi, and Luis Wong

1. Design effort

The design efforts spanned two weeks, with each member dedicating 20 hours of work to advancing the design. The breakdown of tasks and time spent on them is shown in the table below:

	Jeremiah Dados		Luis Wong		Dmytro Stavskyi	
Week	Task	Time spent (hours)	Tasks	Time spent (hours)	Tasks	Time spent (hours)
12/10/2025 - 18/10/2025	Researching HF PA solutions (literature review)	3	Researching HF PA solutions (literature review)	5	Researching EMG signals and how they are usually processed and captured	3
	HF PA simulations: class- A	3	HF PA simulations: Class-A	2	Researching how ADS1299 and the MCU (MSPM0C1104) can interact with each other (SPI and pin connections)	3
	Researching MCU options and creating MCU wiring diagram	4	Creating and presenting slideshow for PAs for weekly meeting w/ professor	3	Setting up Altium, examining the Altium schematics of the original prototype and setting up the library for the ADS1299 chip	4
19/10/2025 - 25/10/2025	Researching HF PA solutions (literature review)	3	Researching HF PA solutions (literature review)	7	Synthesizing information for the and creating the weekly stakeholder presentation about the ADS characteristics	4
	PA circuit topology design	3	PA circuit topology design	3	Researching the use of ADS1299 in EMG and other signal processing systems	4
	MCU circuit topology design	1	HF PA simulations: Class-E	3	Preliminary EMG Recording circuit design	2
	Preliminary PCB design	3				

Table 1: Work breakdown and time commitments per team member

a. System architecture design (Jeremiah)

Overview

The following diagram shows the system architecture we decided on:

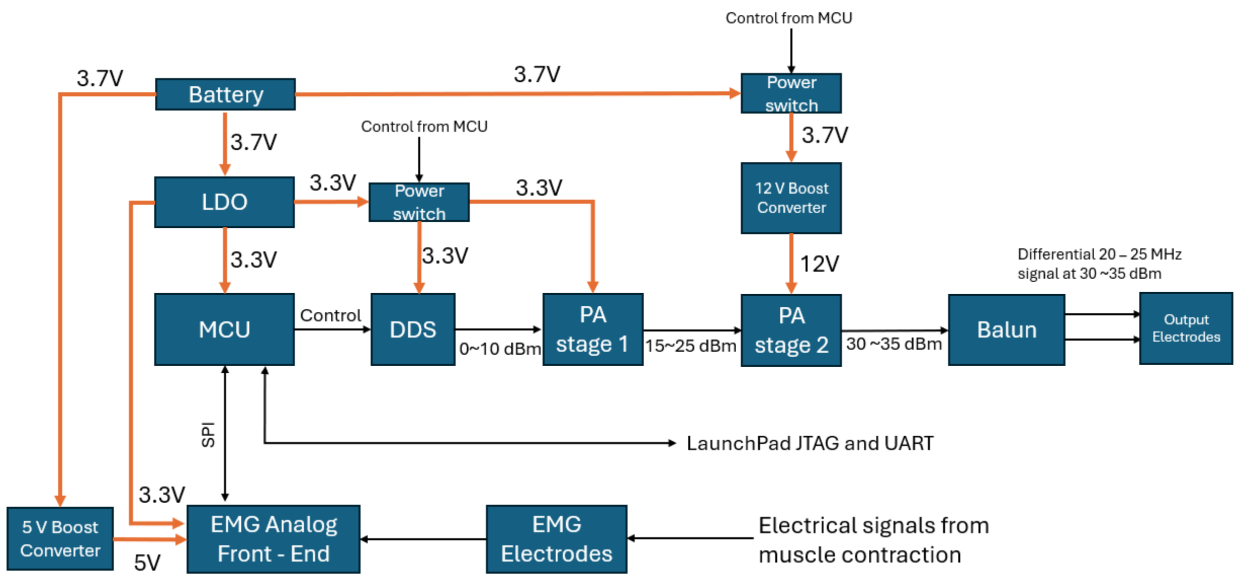


Figure 1: System architecture.

The design constraints received from Dr. Khalifa were the following:

- The device has to use a thin 3.7 V Li-ion battery and be able to withstand at least several hours of continuous operation without charging.
- The output signal will be differential, at 20 - 25 MHz and pulsed at 1 ms / s, with at least 1W and preferably 2W output power.
- The frequency needs to be software-adjustable.
- The output power needs to be software or hardware-adjustable (potentiometer/switch).
- The PCB cannot exceed 70 mm x 50 mm dimensions.

Therefore, all the internal system architecture components needed to be carefully chosen by us based on the literature review and previous experience. All part searches were performed using Digikey and Mouser Electronics searches.

RF Signal Generation

Review of electronic RF signal generator ICs available on the market revealed that a DDS would be the best solution based on how easy it is to digitally tune its output frequency. Other options were rejected for the following reasons:

- VCO: lower output power (below 0 dBm), harder frequency control, fewer options available in the HF band.
- Crystal oscillators: no frequency control, not designed for RF signal generation.

After a careful search of available DDS options, **AD9850** was chosen due to its ability to deliver signals up to 40 MHz, with theoretical output power up to 13 dBm:

AD9850–SPECIFICATIONS (V_S = 5 V ± 5% except as noted, R_{SET} = 3.9 kΩ)

Parameter	Temp	Test Level	AD9850BRS Min	Typ	Max	Unit
CLOCK INPUT CHARACTERISTICS						
Frequency Range						
5 V Supply	Full	IV	1		125	MHz
3.3 V Supply	Full	IV	1		110	MHz
Pulse Width High/Low						
5 V Supply	25°C	IV	3.2			ns
3.3 V Supply	25°C	IV	4.1			ns
DAC OUTPUT CHARACTERISTICS						
Full-Scale Output Current						
R_{SET} = 3.9 kΩ	25°C	V		10.24		mA
R_{SET} = 1.95 kΩ	25°C	V		20.48		mA
Gain Error	25°C	I	−10		+10	% FS
Gain Temperature Coefficient	Full	V		150		ppm/°C
Output Offset	25°C	I			10	µA
POWER SUPPLY (A_{OUT} = 1/3 CLKIN)						
$+V_S$ Current @						
62.5 MHz Clock, 3.3 V Supply	Full	VI		30	48	mA
110 MHz Clock, 3.3 V Supply	Full	VI		47	60	mA
62.5 MHz Clock, 5 V Supply	Full	VI		44	64	mA
125 MHz Clock, 5 V Supply	Full	VI		76	96	mA
P_{DISS} @						
62.5 MHz Clock, 3.3 V Supply	Full	VI		100	160	mW
110 MHz Clock, 3.3 V Supply	Full	VI		155	200	mW
62.5 MHz Clock, 5 V Supply	Full	VI		220	320	mW
125 MHz Clock, 5 V Supply	Full	VI		380	480	mW
P_{DISS} Power-Down Mode						
5 V Supply	Full	V		30		mW
3.3 V Supply	Full	V		10		mW

*Tested by measuring output duty cycle variation.
Specifications subject to change without notice.

Figure 2: DDS output power, voltage supply, and power consumption characteristics. Source: Analog Devices (https://www.analog.com/media/en/technical-documentation/data-sheets/ad9850.pdf)

The output power was calculated based on the maximum output current and a 50-ohm load:

$$P = I^2R = (20.48\, mA)^2 * 50\, ohm = 0.021\, W = 13.22\, dBm$$

The power consumption of around 60 mA would significantly shorten the battery life if the DDS were powered on all the time. Because of that, a power switch will be used by the microcontroller to turn off the DDS during the periods of time when power transmission is not needed.

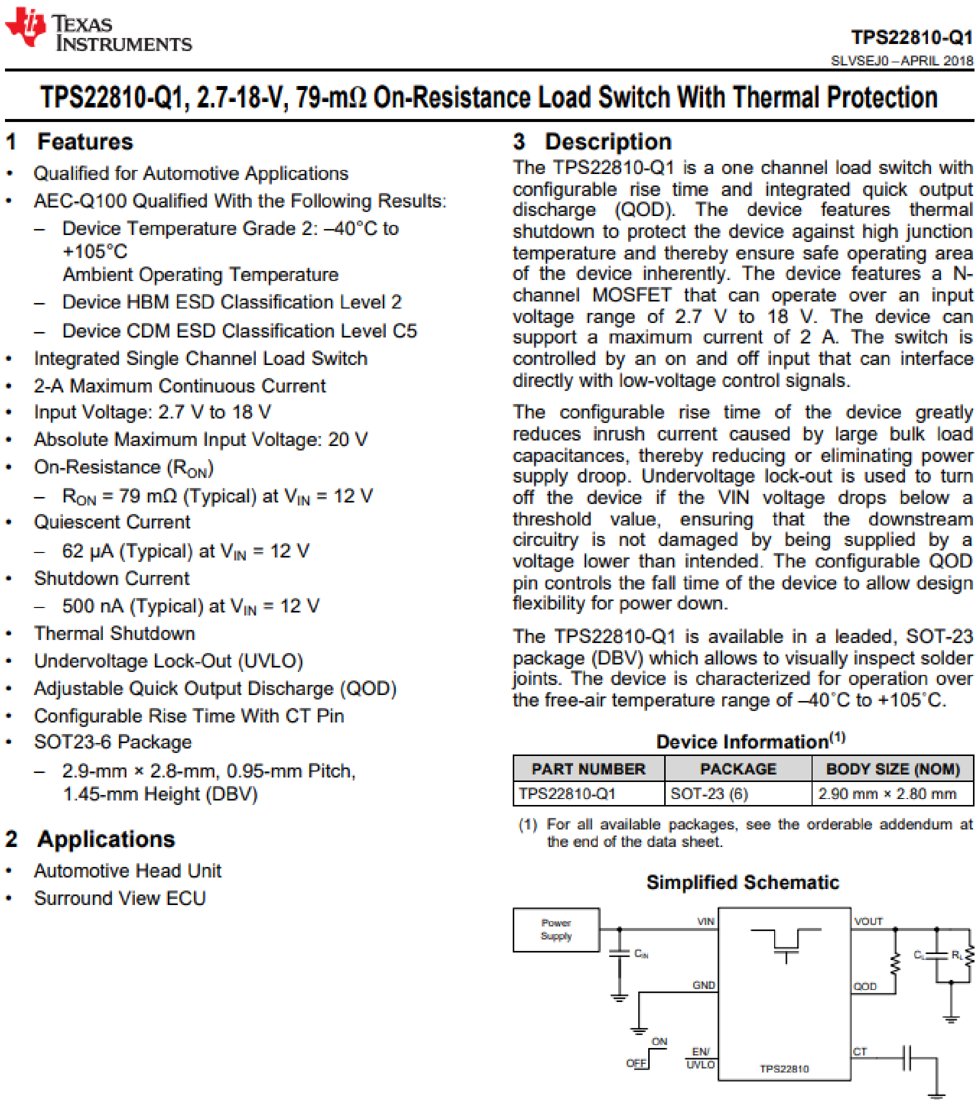
Power switching

The part choice criteria for the power switch were as follows:

- 0 ~ 12 V input and output voltages

- >2A output current
- Simple to use

After a Digikey and Mouser search, TPS22810-Q1 was chosen as the most promising candidate based on its datasheet specs combined with the simplicity of use:



TEXAS INSTRUMENTS

TPS22810-Q1
SLVSEJ0 – APRIL 2018

TPS22810-Q1, 2.7-18-V, 79-mΩ On-Resistance Load Switch With Thermal Protection

1 Features

- Qualified for Automotive Applications
- AEC-Q100 Qualified With the Following Results:
  - Device Temperature Grade 2: –40°C to +105°C Ambient Operating Temperature
  - Device HBM ESD Classification Level 2
  - Device CDM ESD Classification Level C5
- Integrated Single Channel Load Switch
- 2-A Maximum Continuous Current
- Input Voltage: 2.7 V to 18 V
- Absolute Maximum Input Voltage: 20 V
- On-Resistance (R_{ON})
  - R_{ON} = 79 mΩ (Typical) at V_{IN} = 12 V
- Quiescent Current
  - 62 µA (Typical) at V_{IN} = 12 V
- Shutdown Current
  - 500 nA (Typical) at V_{IN} = 12 V
- Thermal Shutdown
- Undervoltage Lock-Out (UVLO)
- Adjustable Quick Output Discharge (QOD)
- Configurable Rise Time With CT Pin
- SOT23-6 Package
  - 2.9-mm × 2.8-mm, 0.95-mm Pitch, 1.45-mm Height (DBV)

2 Applications

- Automotive Head Unit
- Surround View ECU

3 Description

The TPS22810-Q1 is a one channel load switch with configurable rise time and integrated quick output discharge (QOD). The device features thermal shutdown to protect the device against high junction temperature and thereby ensure safe operating area of the device inherently. The device features a N-channel MOSFET that can operate over an input voltage range of 2.7 V to 18 V. The device can support a maximum current of 2 A. The switch is controlled by an on and off input that can interface directly with low-voltage control signals.

The configurable rise time of the device greatly reduces inrush current caused by large bulk load capacitances, thereby reducing or eliminating power supply droop. Undervoltage lock-out is used to turn off the device if the VIN voltage drops below a threshold value, ensuring that the downstream circuitry is not damaged by being supplied by a voltage lower than intended. The configurable QOD pin controls the fall time of the device to allow design flexibility for power down.

The TPS22810-Q1 is available in a leaded, SOT-23 package (DBV) which allows to visually inspect solder joints. The device is characterized for operation over the free-air temperature range of –40˚C to +105˚C.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
TPS22810-Q1	SOT-23 (6)	2.90 mm × 2.80 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Simplified Schematic

Figure 3: The power switch chosen for the design. Source: Texas Instruments (https://www.ti.com/product/TPS22810-Q1/part-details/TPS22810TDBVRQ1)

Power amplification

It is understood that the real output power of the DDS on the PCB might be less than the theoretical value. Because of that, the PA was designed to work with input signals as low as 5 dBm. Different topologies, such as Class-A and Class-E, were considered in the design process. The design is described in-depth in section 1b of this document.

EMG recording
The choice of the AFE (analog front-end) chip was a critically important part of the design. The programming interface, sampling rate, amplifier bandwidth, and other signal processing factors were considered. The design is described in depth in section 1c of this document.

Microcontroller
When trying to find a suitable microcontroller for this embedded system, the main criteria of the search were the following:

- Minimal external circuitry needed for programming and operation.
- Minimal part cost.
- Minimal current consumption.
- Sufficient number of GPIO pins to support DDS programming, communication with the AFE, and control of power switches.

The search was conducted among Texas Instruments ARM-based microcontrollers. That family of devices was chosen due to our familiarity with TI firmware development environments and evaluation platforms. Ultimately, the **MSPM0C1104** device was chosen due to its small footprint and minimal external circuits required, as shown on its evaluation board schematic:

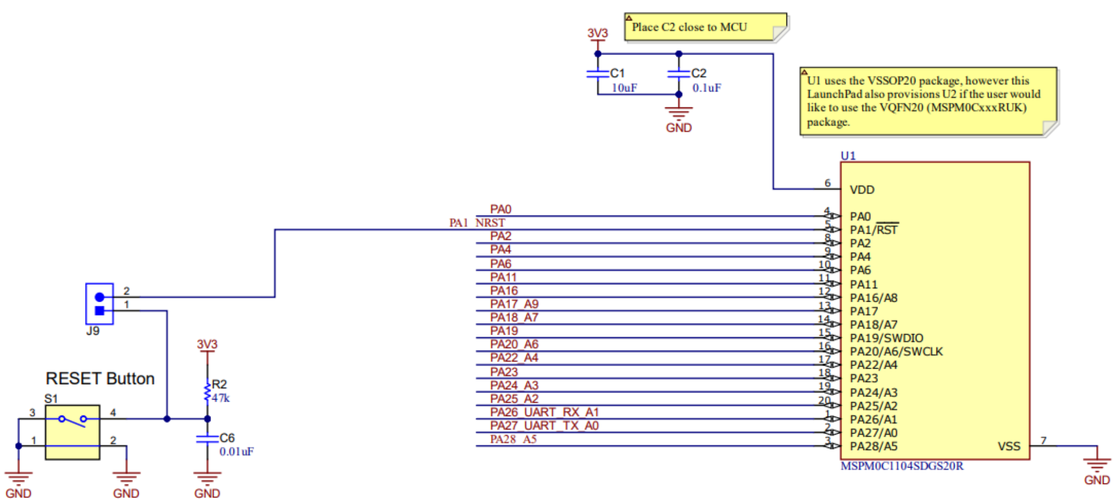


Figure 4: MSPM0C1104 requires only 4 external passive components to operate. Source: Texas Instruments (https://www.ti.com/tool/LP-MSPM0C1104#design-files)

Furthermore, only 3 pins (SWCLK, SWDIO, NRST) with no external ICs on board would be needed to program the device using a LaunchPad.
The microcontroller datasheet was used to verify that it can be powered directly from the battery or through a 3.3V LDO (preferred), and the expected current consumption is below 3 mA:

7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	NOM	MAX	UNIT
VDD	Supply voltage [2]	1.62 [3]		3.6	V
C_{VDD}	Capacitor placed between VDD and VSS [1]		10		uF
T_A	Ambient temperature	–40		125	°C
T_J	Max junction temperature			130	°C
f_{MCLK}	MCLK, CPUCLK, ULPCLK frequency with 0 flash wait states			24	MHz

7.5.1 RUN/SLEEP Modes

VDD=3.3V. All inputs tied to 0V or VDD. Outputs do not source or sink any current. All peripherals are disabled.

PARAMETER		MCLK	-40°C		25°C		85°C		125°C		UNIT
			TYP	MAX	TYP	MAX	TYP	MAX	TYP	MAX	
RUN Mode											
IDD_{RUN}	MCLK=SYSOSC, While(1), execute from flash	24MHz	2.06	2.20	2.08	2.35	2.09	2.40	2.21	2.45	mA
IDD_{RUN}, per MHz	MCLK=SYSOSC, While(1), execute from flash	24MHz	86	92	87	98	87	100	92	102	uA/Mhz
SLEEP Mode											
IDD_{SLEEP}	MCLK=SYSOSC, CPU is halted	24MHz	1115	1256	1132	1268	1149	1380	1214	1370	uA

Figure 5: Electrical characteristics of MSPM0C1104. Source: Texas Instruments (https://www.ti.com/tool/LP-MSPM0C1104#design-files)

After choosing the microcontroller, its connections with other ICs were planned out to make sure there are enough GPIO pins for the design:

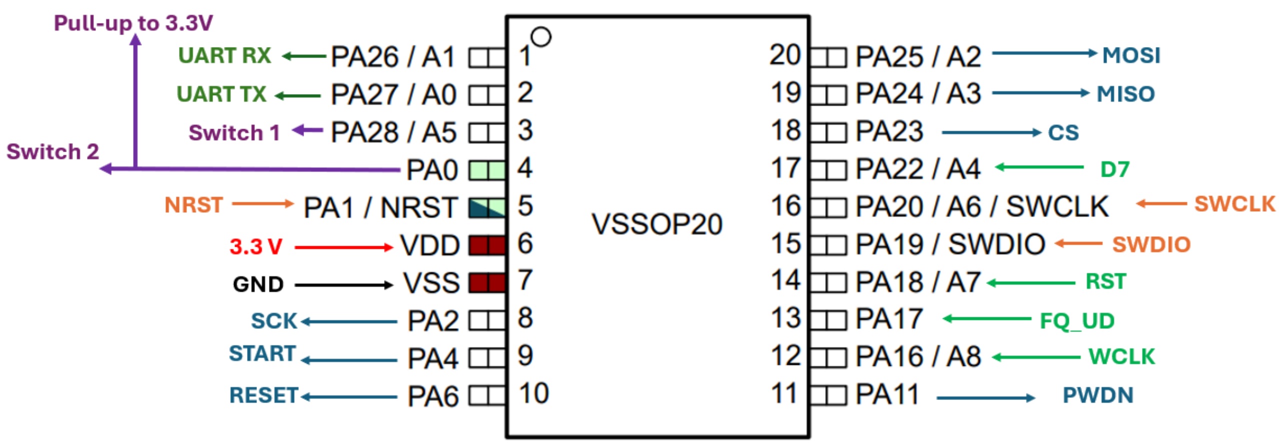


Figure 6: Pin assignments on the microcontroller

The dark green pins are used for UART communications, the purple are used for power switching, the orange are used for JTAG / flashing, the blue are used for the EMG chip, and the bright green are used for the DDS.

Power supplies

The requirements for the power circuit of the device were as follows:

- At least 2A output current limit for all power ICs
- LDO needed to drop 3.7V to 3.3V for the MCU and EMG AFE

- Boost converters needed to obtain 5.0 V and 12.0 V from the battery for the EMG AFE and the power amplifier.

The following ICs were chosen to meet the requirements:


life.augmented

LD39200

Datasheet

2 A high PSRR ultra low drop linear regulator with reverse current protection

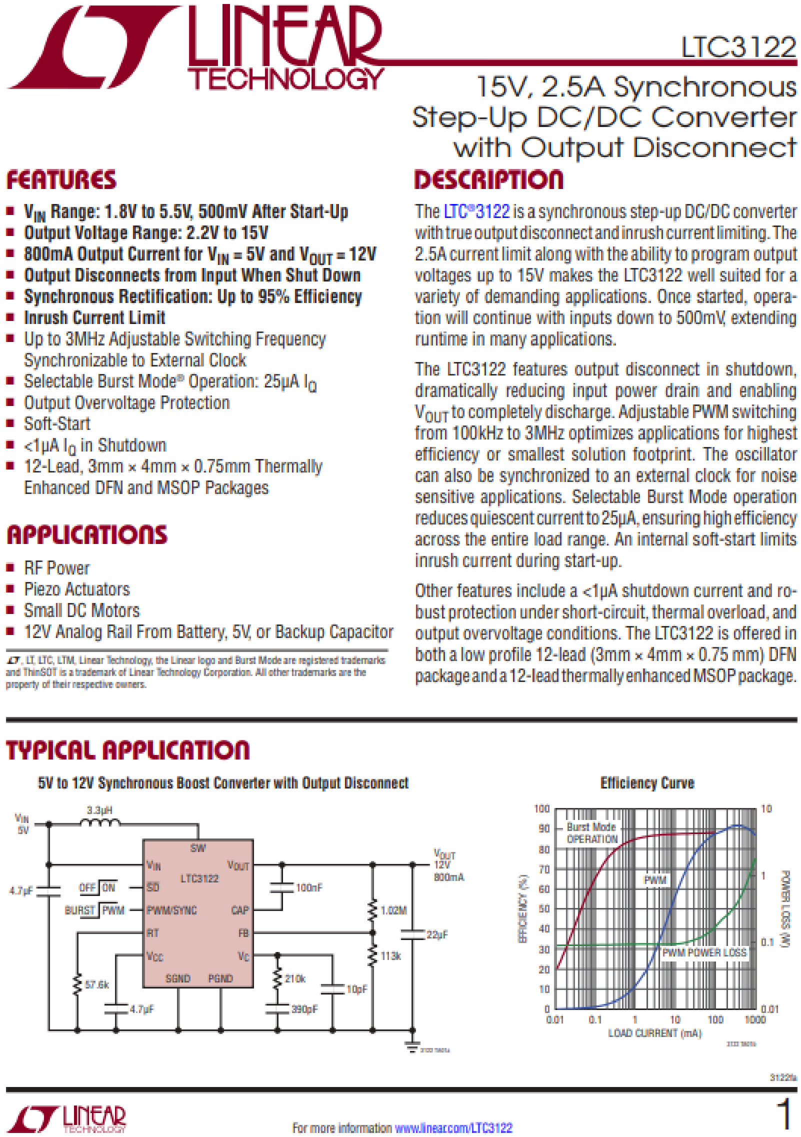
DFN8 (4 x 4 mm)

DFN6 (3 x 3 mm)

Features

- Input voltage from 1.25 V to 6.0 V
- Ultra low drop: 130 mV (typ.) at 2 A load
- 1 % output accuracy at 25 °C, 2 % in full temperature range
- High PSRR: 70 dB at 1 kHz
- Reverse current protection
- 2 A guaranteed output current
- Available in fixed and adjustable output voltage version from 0.5 V with 100 mV step
- Power Good
- Internal current and thermal limit
- Operating junction temperature range: -40 °C to 125 °C
- DFN6 (3 x 3 mm) and DFN8 (4 x 4 mm) packages

Figure 7: LDO chosen for the design. Source: STMicroelectronics (https://www.st.com/content/ccc/resource/technical/document/datasheet/4b/e6/2e/a4/16/04/46/9d/DM00102135.pdf/files/DM00102135.pdf/jcr:content/translations/en.DM00102135.pdf)



LTC3122

15V, 2.5A Synchronous Step-Up DC/DC Converter with Output Disconnect

FEATURES

- **V_{IN} Range: 1.8V to 5.5V, 500mV After Start-Up**
- **Output Voltage Range: 2.2V to 15V**
- **800mA Output Current for V_{IN} = 5V and V_{OUT} = 12V**
- **Output Disconnects from Input When Shut Down**
- **Synchronous Rectification: Up to 95% Efficiency**
- **Inrush Current Limit**
- Up to 3MHz Adjustable Switching Frequency Synchronizable to External Clock
- Selectable Burst Mode® Operation: 25μA I_Q
- Output Overvoltage Protection
- Soft-Start
- <1μA I_Q in Shutdown
- 12-Lead, 3mm × 4mm × 0.75mm Thermally Enhanced DFN and MSOP Packages

APPLICATIONS

- RF Power
- Piezo Actuators
- Small DC Motors
- 12V Analog Rail From Battery, 5V, or Backup Capacitor

DESCRIPTION

The LTC®3122 is a synchronous step-up DC/DC converter with true output disconnect and inrush current limiting. The 2.5A current limit along with the ability to program output voltages up to 15V makes the LTC3122 well suited for a variety of demanding applications. Once started, operation will continue with inputs down to 500mV, extending runtime in many applications.

The LTC3122 features output disconnect in shutdown, dramatically reducing input power drain and enabling V_{OUT} to completely discharge. Adjustable PWM switching from 100kHz to 3MHz optimizes applications for highest efficiency or smallest solution footprint. The oscillator can also be synchronized to an external clock for noise sensitive applications. Selectable Burst Mode operation reduces quiescent current to 25μA, ensuring high efficiency across the entire load range. An internal soft-start limits inrush current during start-up.

Other features include a <1μA shutdown current and robust protection under short-circuit, thermal overload, and output overvoltage conditions. The LTC3122 is offered in both a low profile 12-lead (3mm × 4mm × 0.75 mm) DFN package and a 12-lead thermally enhanced MSOP package.

TYPICAL APPLICATION

Figure 8: Boost converter chosen for the design. Source: STMicroelectronics (https://www.analog.com/en/products/ltc3122.html)

The LTC3122 boost converter will be used to generate both 5.0 V and 12.0 V outputs, because two instances of the chip can be placed on the PCB, each with a different voltage divider circuit setting a different output voltage.

b. PA design and simulation (Luis)

One of the main goals of the device is to present a wireless neurostimulation platform powered via Differential Tissue Coupled Powering (DTCP). This is meant to be achieved through the use of high-frequency (~ MHz) currents through conductive tissue using epidermal electrodes positioned near the implant. To this end, we need a reliable way to not only produce a high-frequency signal but also ensure that the signal has sufficient power to run the implanted devices.

I was tasked with researching and developing the PA we will need. As someone who lacks extensive experience and knowledge in designing not only PAs but also analog devices, I needed to do a lot of research in order to sufficiently propose designs to the professor for our weekly meetings.

RF (radio frequency) is a broader term, referring to any frequency used for radio communications (from e.g., tens of kHz up through MHz, GHz, etc.). While HF (high frequency) refers to a specific band within the RF umbrella between 3–30 MHz. With this information in mind, I was able to learn that what we needed to look for was information regarding RF and HF amplifiers.

Upon researching, I made use of this [1] as a primer on RF PAs due to its target audience being that of beginners and hobbyists. This allowed me to gain sufficient knowledge in order to learn about the different major amplifier classes.

Class	Conduction Angle	Power Efficiency	Linearity
A	360° (always on)	~25-35%	Highest
B	180°	~65%	Moderate
AB	180-360°	~50-70%	Better than AB
C	>360°	~60-80%	Poor
D	Switch	~80-90%	Poor
E	Switch	~80-90%	Moderate
F	Switch + harmonic tuning	~85-90%	Moderate

Figure 9: A table summarizing the characteristics of the different amplifier classes [1],[3],[4].

- Linearity: refers to an amplifier's ability to produce an output that is a faithful, scaled reproduction of its input, without unwanted distortion.

As you can see from the table, there are two main categories of amplifiers: the linear PAs (Class A, B, AB, C) and the switching PAs (Class D, E, F). Linear amplifier's active device(s) operate in their linear (saturation) region (i.e., the output current/voltage is proportional to the input over the waveform). These are designed to preserve the waveform (amplitude and phase) and thus minimize distortion. However, linear operation means the device often has voltage across it and current through it concurrently, which leads to power dissipation (heat) and lower efficiency. A switching PA uses devices that are either fully ON or fully OFF (instead of continuously in between). Because when the device is ON, the voltage across it is low (so voltage × current ≈ small), and when it is OFF, the current through it is low (so again little dissipation), the overall power wasted is much lower, which leads to higher efficiency.

The trade‑off: switching amplifiers often require special output networks, harmonic filtering, or tuned loads, and may not support arbitrary linear modulation with high fidelity unless compensated. Linear PAs support better fidelity/linearity, but at the cost of efficiency and heat.

In our case, we do not care too much about the linearity of the waveform, just that it has a certain power (30-35 dBm) and it has a frequency ranging from 20-25 MHz. We also care about power efficiency and consumption, as our goal for this device is to run it for hours on a single battery charge. For this reason, switch PAs fit our purposes better. To that end, we decided on a Class E power amplifier design.

Why not Class D: Class D is a switching mode PA with high efficiency, but in RF applications, it has limitations: switching losses, parasitic capacitances, and device speed become more challenging at high RF frequencies. In general, it becomes very difficult to design around the limitations of Class D in HF environments.

Why not Class F: Class F uses harmonic tuning and can reach very high efficiencies, but it can add design complexity (harmonic resonators, load network complexity) and may be less attractive if we need a more straightforward PA. We would like the PCB to be relatively small; even if there was efficiency to be gained, the time and space it would take to reach that efficiency is not worth it.

Why Class E: Class E is specifically well-suited for RF power amplification in switching mode. Key reasons:

- It uses a switching element plus a tuned reactive network so that switching transitions occur under near-zero-voltage or zero-current conditions, which minimizes switching losses
- It achieves very high efficiency, which in RF power is critical (less wasted heat, smaller thermal budget, smaller size, better reliability).
- Its design is simpler (relatively) for RF switching PAs.

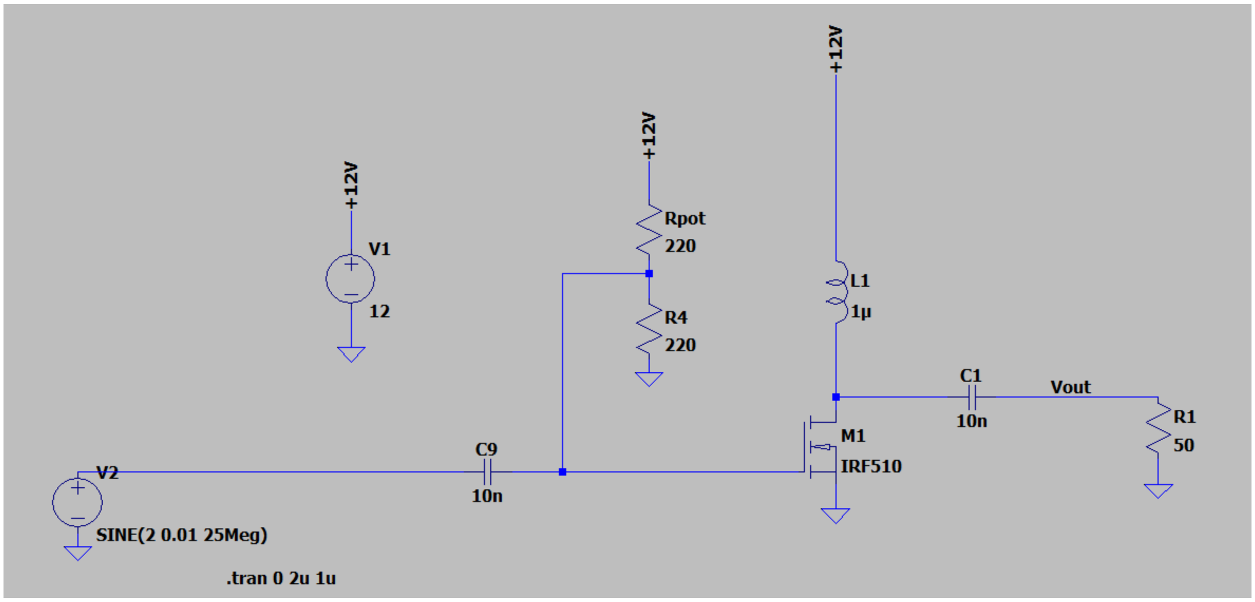


Figure 10: When we began the design of the PA, we started with the simplest one, a Class-A PA (this was before research).

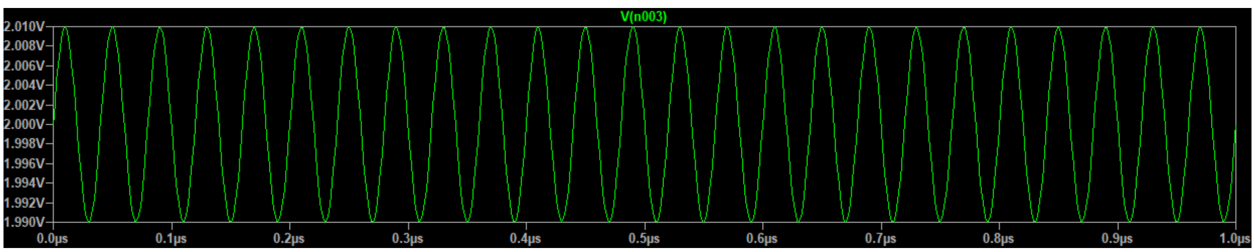


Figure 11: Input signal simulating what would be output from the DDS.

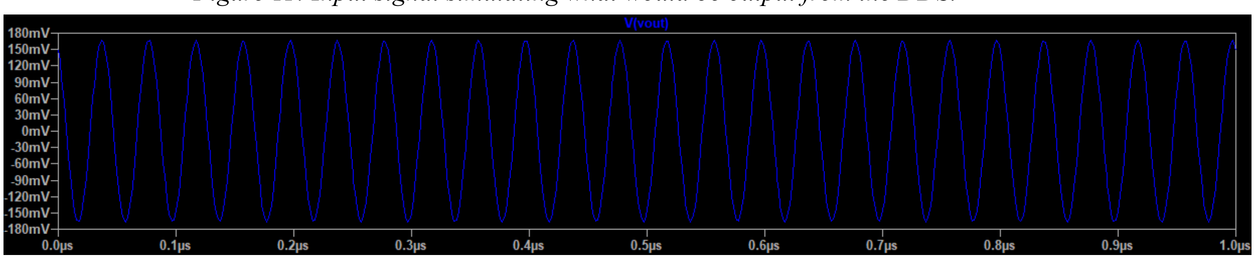


Figure 12: Output signal showing that the signal was amplified.

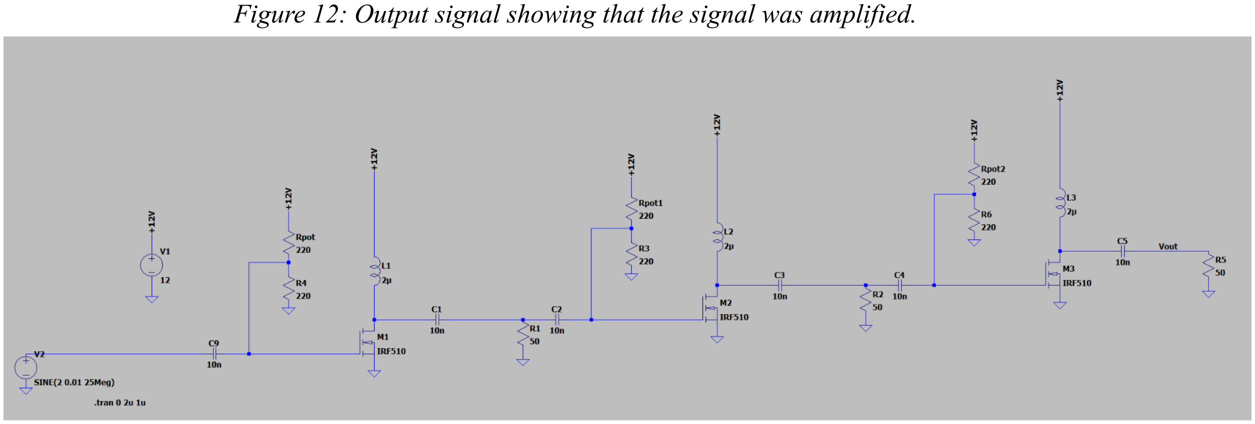


Figure 13: Gain was insufficient, so we decided to put three Class-A amplifiers in series.

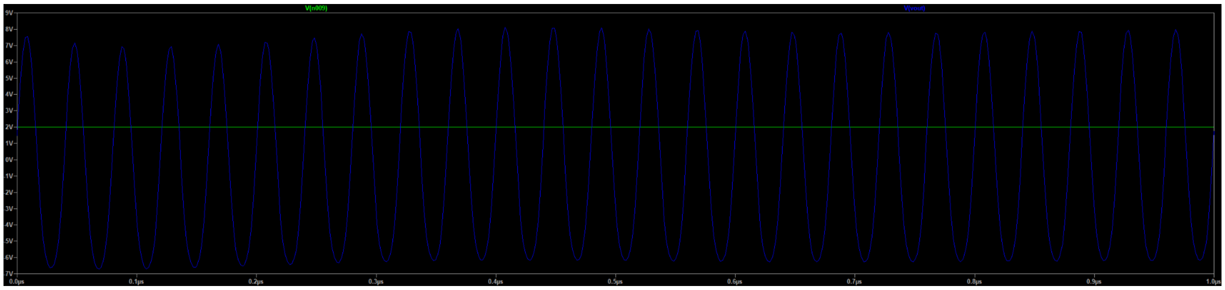


Figure 14: The Signal shows amplification resulting in a signal of ~14 Vpp, which is about 27 dBm. The amplitude of the signal oscillated in amplitude but with some tuning; it was dampened somewhat, but not enough to be ideal. If we want to reach the desired power, it would require too many transistors.

For this design, we drew inspiration from [2] for parts/layout and utilized a calculator (https://people.physics.anu.edu.au/~dxt103/calculators/class-e.php) based on formulas found in [5].

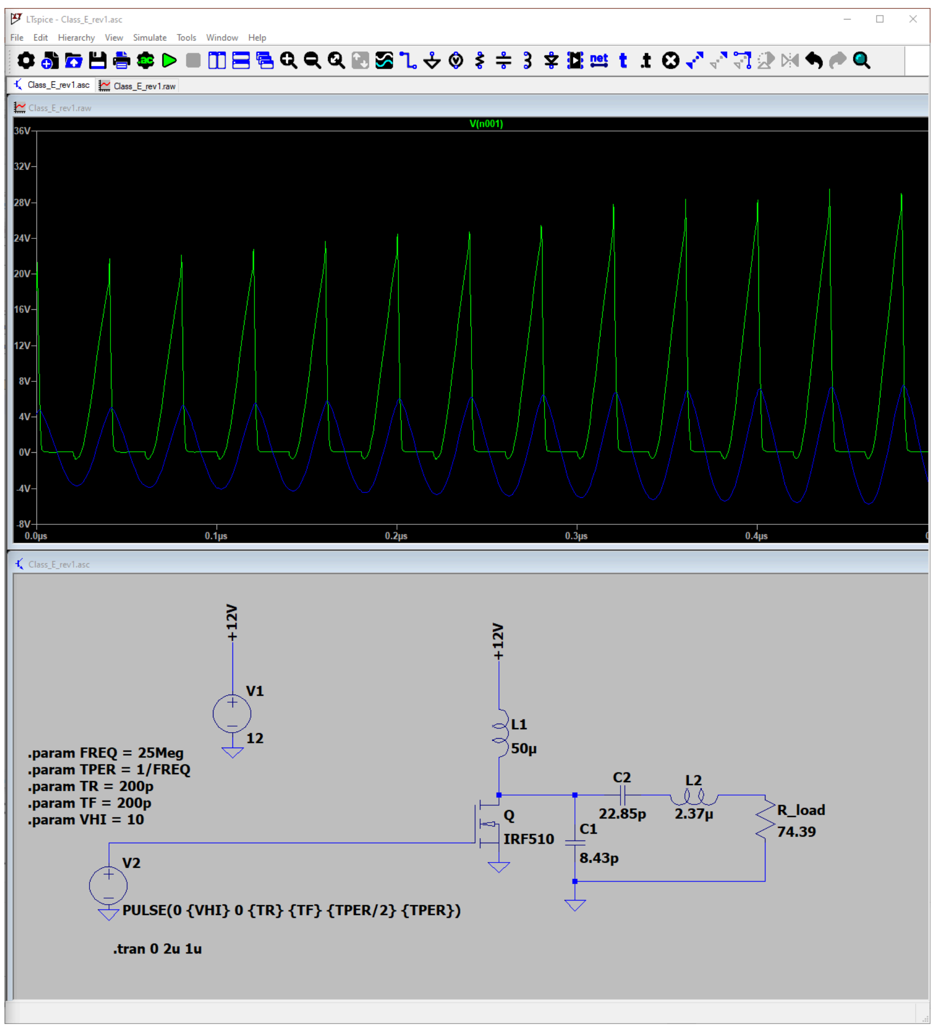


Figure 15: Initial attempt at designing and simulating the Class E PA. Expected a Vpp of 24 but got a Vpp of 8 instead.

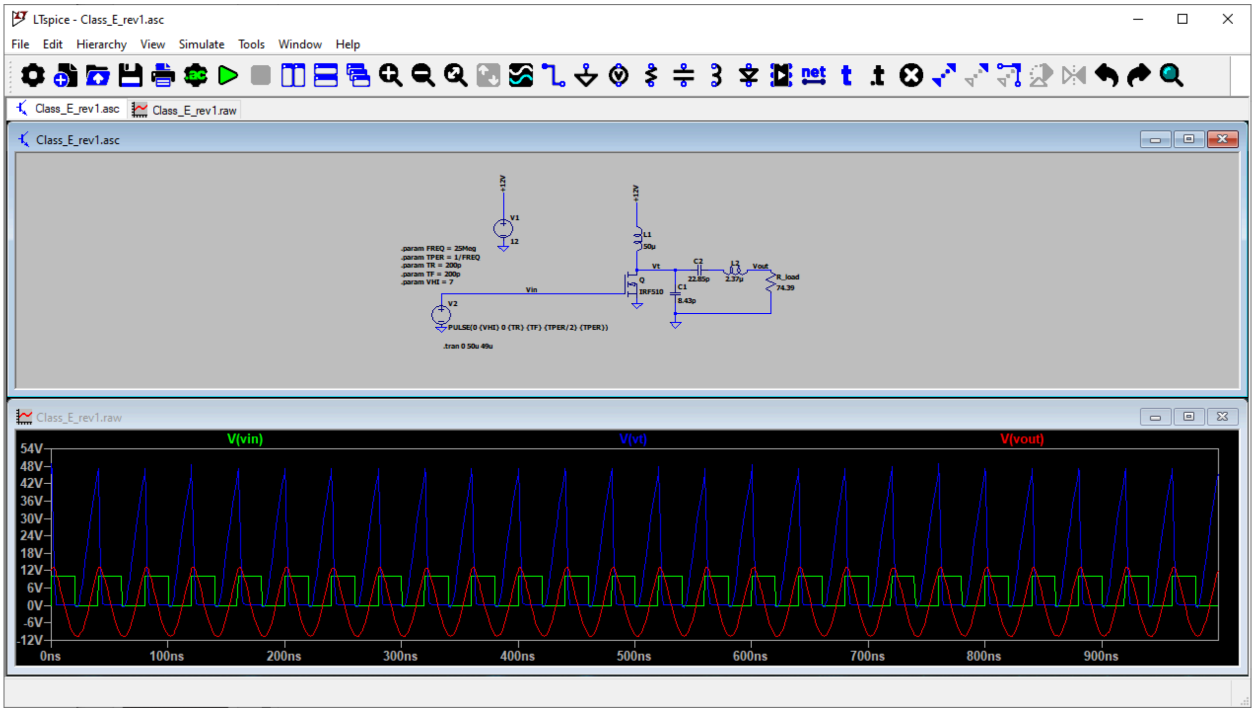


Figure 16: After making the simulation wait 40 microseconds, it seems like the result matched the expectation. The circuit seems to need a little time to warm up.

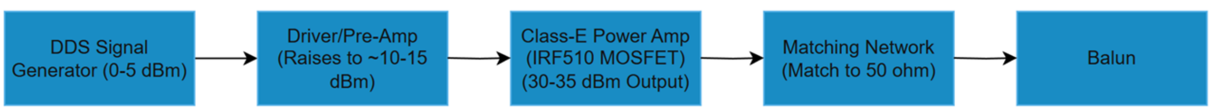


Figure 17: High-level view of PAs' planned design.

Here, we make use of a Balun instead of a differential PA, as it reduces the complexity of the circuit. This allows us to keep a small footprint.

Quiescent current is the current drawn by a circuit when it is powered on but not delivering any signal or load power. In switch PAs, quiescent current is very low (about 10-15 pA) or nearly zero because the transistor is off most of the time.

A power switch (like the MP5000ADQ-LF-P) is used in order to turn off the 12V boost converter, which will turn off the PA, which will further prevent any power draw from the Class E PA. A similar setup will also be utilized in the Driver/Pre-Amp and DDS stage.

A concern with the MOSFET is that its package size is too large; we can go for a smaller, more efficient option in the form of the GaN-FET, though there are concerns that this may make debugging the board more difficult, as the pins are less accessible. A good argument for using it can be found in [6]. At 13–27 MHz, device output capacitance and gate charge dominate switching loss and ringing. The paper explicitly selects an enhancement-mode GaN FET (EPC2037) for its very low capacitances and pairs it with a fast driver to achieve clean ZVS waveforms and high efficiency. This directly underpins the choice of GaN over IRF-class Si parts.

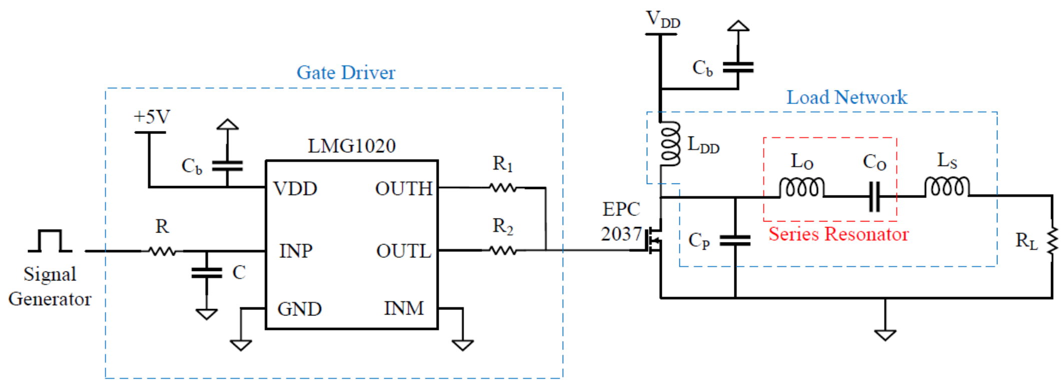


Figure 18: A signal generator (clock oscillator) feeds a pulse to the LMG1020 gate driver integrated with the Class E PA circuit, a good example layout for how a gate driver would drive the transistor in the next stage [6].

c. EMG design

The signal we intended to capture and record with this system is of the EMG (Electromyography) type. Specifically, we are capturing and recording the electrical activity of the animal muscles (e.g., muscle contraction).
The frequency of this type of signal is up to 500 Hz, and the amplitude ranges up to 10 mV [7].

The main criteria for the chip include minimal power consumption (since it’s a battery-powered system) and the ability to properly capture and process EMG signals (that is, adequate amplifier gain and bandwidth, sampling rate, etc.).

The AFE chip we chose for this part of the system is ADS1299-4 (4 channels).

EMG processing systems can be divided into four stages: sensing, amplifying, filtering, and ADC.

The sensing is done through analog inputs, which will be connected to electrodes (which in turn are needles connected to the animal).

7.5 Electrical Characteristics

Minimum and maximum specifications apply from T_A = –40°C to 85°C. Typical specifications are at T_A = +25°C. All specifications are at AVDD – AVSS = 5 V, DVDD = 3.3 V, V_{REF} = 4.5 V, external f_{CLK} = 2.048 MHz, data rate = 250 SPS, and gain = 12 (unless otherwise noted)

AC CHANNEL PERFORMANCE						
CMRR	Common-mode rejection ratio	f_{CM} = 50 Hz and 60 Hz (2)	–110	–120		dB
PSRR	Power-supply rejection ratio	f_{PS} = 50 Hz and 60 Hz		96		dB
	Crosstalk	f_{IN} = 50 Hz and 60 Hz		–110		dB
SNR	Signal-to-noise ratio	V_{IN} = –2 dBFs, f_{IN} = 10-Hz input, gain = 12		121		dB
THD	Total harmonic distortion	V_{IN} = –0.5 dBFs, f_{IN} = 10 Hz		–99		dB

Figure 19: ADS1299 Analog Channel Performance. Source : Texas Instruments (https://www.ti.com/product/ADS1299)

The device has strong general characteristics for the performance of AC channels, like low crosstalk (signals interfering with each other; specifically -110 dB for 50 Hz and 60 Hz input signals), high (im absolute magnitude) common-mode rejection ratio (CMRR; -120 dB for a particular CM frequencies) and high signal-to-noise ratio (SNR; 121 dB for a particular input),

ADS1299-x Low-Noise, 4-, 6-, 8-Channel, 24-Bit, Analog-to-Digital Converter for EEG and Biopotential Measurements

1 Features

- Up to Eight Low-Noise PGAs and Eight High-Resolution Simultaneous-Sampling ADCs
- Input-Referred Noise: 1 μV_{PP} (70-Hz BW)
- Input Bias Current: 300 pA
- Data Rate: 250 SPS to 16 kSPS
- CMRR: –110 dB
- Programmable Gain: 1, 2, 4, 6, 8, 12, or 24
- Unipolar or Bipolar Supplies:
  - Analog: 4.75 V to 5.25 V
  - Digital: 1.8 V to 3.6 V
- Built-In Bias Drive Amplifier, Lead-Off Detection, Test Signals
- Built-In Oscillator
- Internal or External Reference
- Flexible Power-Down, Standby Mode
- Pin-Compatible with the ADS129x
- SPI-Compatible Serial Interface
- Operating Temperature Range: –40°C to +85°C

2 Applications

- Medical Instrumentation Including:
  - Electroencephalogram (EEG) Study
  - Fetal Electrocardiography (ECG)
  - Sleep Study Monitor
  - Bispectral Index (BIS)
  - Evoked Audio Potential (EAP)

3 Description

The ADS1299-4, ADS1299-6, and ADS1299 devices are a family of four-, six-, and eight-channel, low-noise, 24-bit, simultaneous-sampling delta-sigma ($\Delta\Sigma$) analog-to-digital converters (ADCs) with a built-in programmable gain amplifier (PGA), internal reference, and an onboard oscillator. The ADS1299-x incorporates all commonly-required features for extracranial electroencephalogram (EEG) and electrocardiography (ECG) applications. With its high levels of integration and exceptional performance, the ADS1299-x enables the creation of scalable medical instrumentation systems at significantly reduced size, power, and overall cost.

The ADS1299-x has a flexible input multiplexer per channel that can be independently connected to the internally-generated signals for test, temperature, and lead-off detection. Additionally, any configuration of input channels can be selected for derivation of the patient bias output signal. Optional SRB pins are available to route a common signal to multiple inputs for a referential montage configuration. The ADS1299-x operates at data rates from 250 SPS to 16 kSPS. Lead-off detection can be implemented internal to the device using an excitation current sink or source.

Multiple ADS1299-4, ADS1299-6, or ADS1299 devices can be cascaded in high channel count systems in a daisy-chain configuration. The ADS1299-x is offered in a TQFP-64 package specified from –40°C to +85°C.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
ADS1299-x	TQFP (64)	10.00 mm × 10.00 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Block Diagram

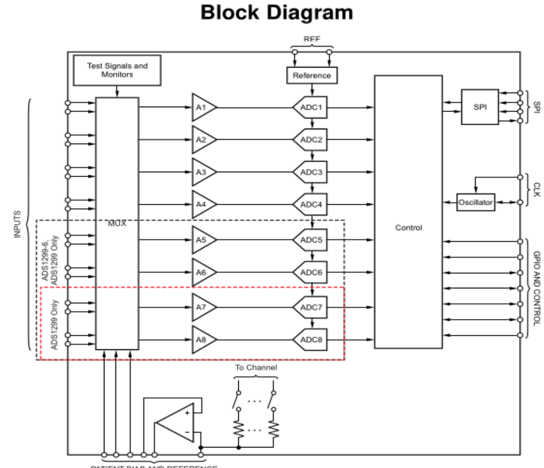


Figure 20: ADS1299 Datasheet. Source : Texas Instruments (https://www.ti.com/product/ADS1299)

For the gain: ADS1299 provides an internal programmable gain amplifier (PGA). At PGA gain of 24x, PGA bandwidth (nominal; at room temperature) is 27 kHz, which is more than enough to properly amplify the EMG signal.

Filtering will be done externally, with some form of a bandwidth filter to decouple low-frequency interferences.

The ADC:

- Resolution: 24 bits
- With an internal reference of 4.5 V, 4.5 V / 2^{24} = 0.268 µV
- Possible sampling rate ranges from 250 SPS (samplings per second) to 16 kSPS (most systems use ~1000 SPS, but we may use less than that)

Power consumption (at 250 SPS):

- The quiescent power consumption of the chip is typically 10 µW.
- The measurement power consumption is typically 22 mW, and the current consumption is typically about 4.60 mA (4.06 mA for analog and 0.54 mA for digital parts)

For power supplies, the analog part of the chip will use 5 V and the digital part will use 3.3 V.

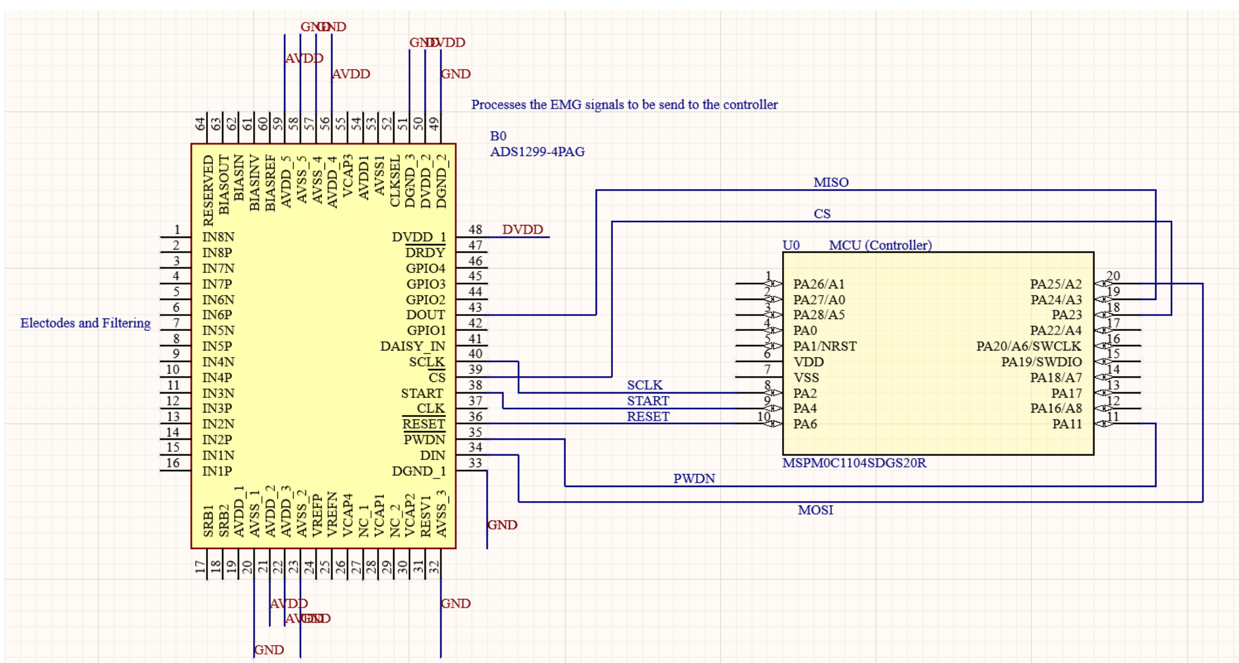


Figure 21: Schematic of ADC1299 pin assignments to our MCU

The chip will be interfaced with the MCU using SPI (serial communication). It can be fully programmed through SPI, though we will connect some specific signals for debugging purposes. The data conversion can be made continuous or one-shot, though the former is more fitting for the purposes of recording.

2. Architectural Elements

a. External Interface

There are two mechanisms of transferring external information to the microcontroller. The first one deals with flashing the device with firmware developed in the Code Composer Studio. To easily program the device from our laptops, we'll use the MSPM0C1104 LaunchPad. The board contains an MSPM0C1104 microcontroller connected to an XDS debugger implemented on an MSP432 device through a jumper header in the middle of the board:

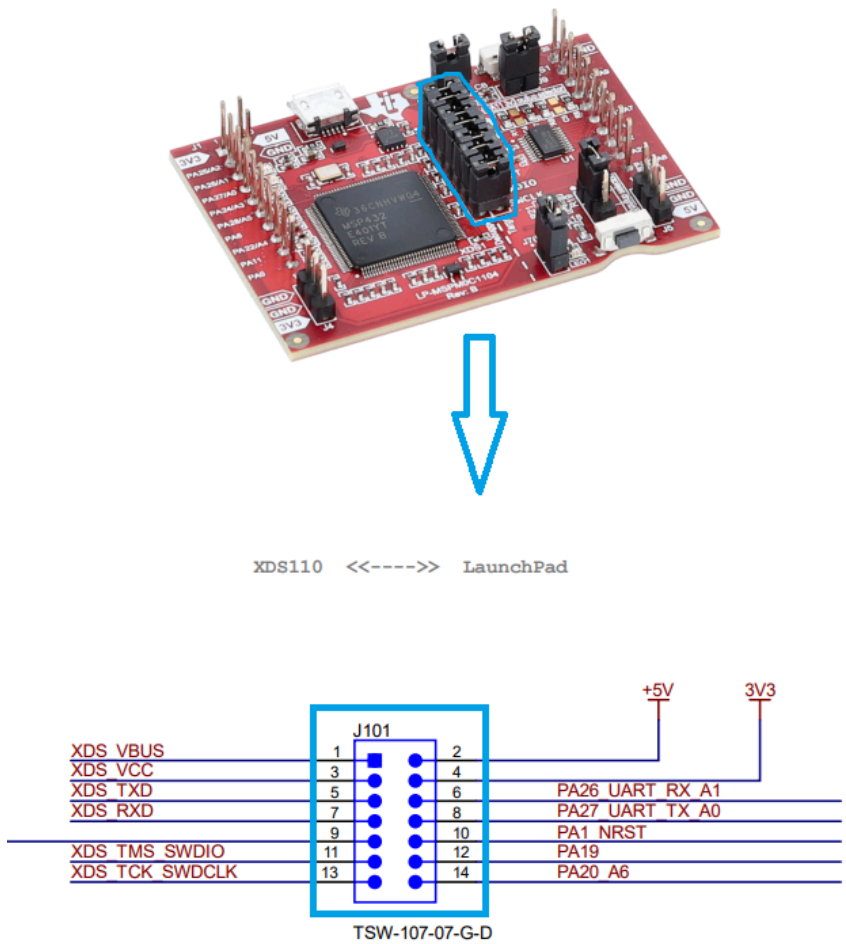


Figure 22: Header connecting the debug/flash circuit to an on-board MSPM0C1104 on the LaunchPad. Source: Texas Instruments (https://www.ti.com/tool/LP-MSPM0C1104#design-files)

We can remove the jumpers from the header and route the debugger circuit to the MSP microcontroller located on our custom PCB instead of the one located on the LaunchPad:

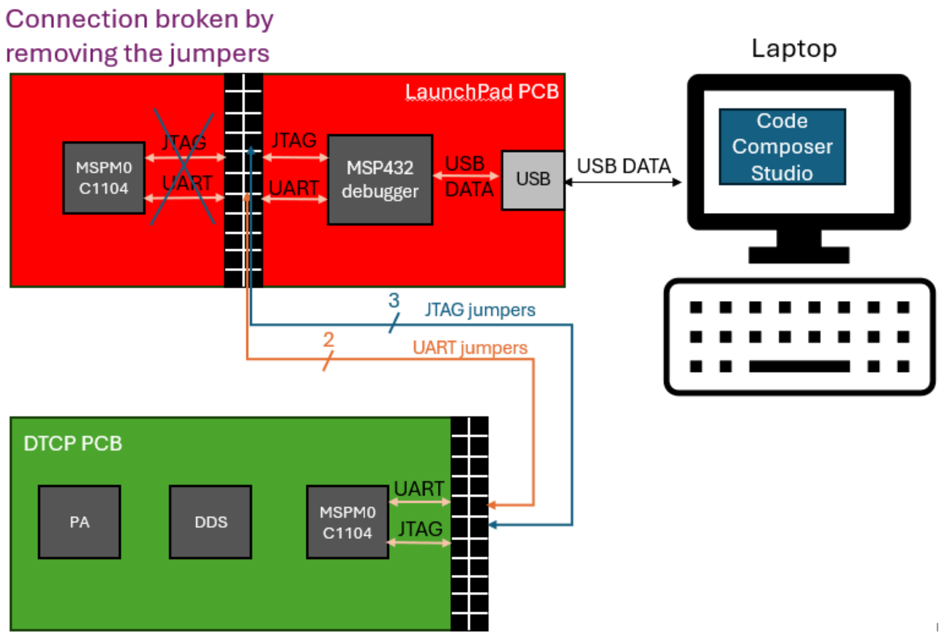


Figure 23: Connections between the firmware developer's laptop, the Launchpad, and the DTCP PCB enabling programming and UART communication between the devices.

This makes the debugger treat the microcontroller on our DTCP board as if it were a part of the LaunchPad, enabling easy flashing and UART communications through the TI-developed interfaces included in the LaunchPad firmware and the Code Composer Studio firmware.

The second interface conveys the EMG voltage information from the muscle via the electrodes to the ADS1299 integrated circuit. The device amplifies the low-voltage analog signals and performs analog-to-digital conversion upon an SPI request from the microcontroller:

9.5.3.9 *RDATA: Read Data*

The RDATA command loads the output shift register with the latest data when not in Read Data Continuous mode. Issue this command after DRDY goes low to read the conversion result. There are no SCLK rate restrictions for this command, and there is no wait time needed for the subsequent commands or data retrieval SCLKs. To retrieve data from the device after the RDATA command is issued, make sure either the START pin is high or the START command is issued. When reading data with the RDATA command, the read operation can overlap the next DRDY occurrence without data corruption. Figure 47 shows the recommended way to use the RDATA command. RDATA is best suited for ECG- and EEG-type systems, where register settings must be read or changed often between conversion cycles.

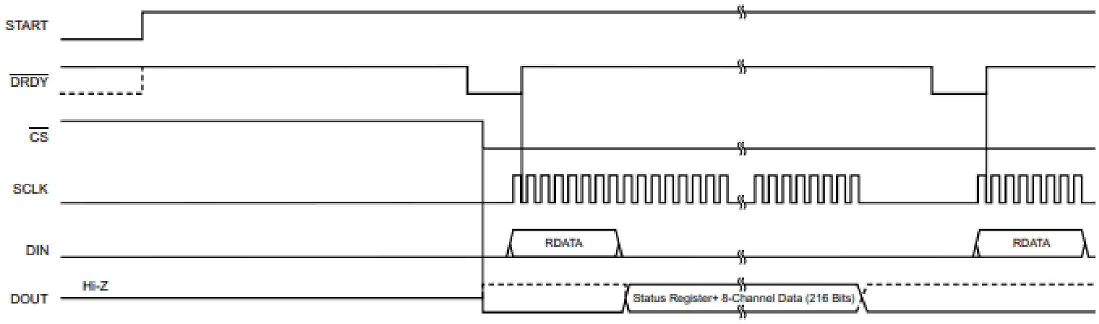


Figure 47. RDATA Usage

Figure 24: A timing diagram of reading data from the AFE using SPI. Source: Texas Instruments (https://www.ti.com/lit/ds/symlink/ads1299.pdf)

The PCB will establish the correct connections between the MSPM0C1104 and ADS1299, while the firmware will ensure the control signals, data, and clock are correctly applied or read at the right times.

b. Persistent State

The following data flow diagram shows how the EMG voltages will be collected from the external interface (electrodes) and stored persistently (in a .txt file on the user's laptop):

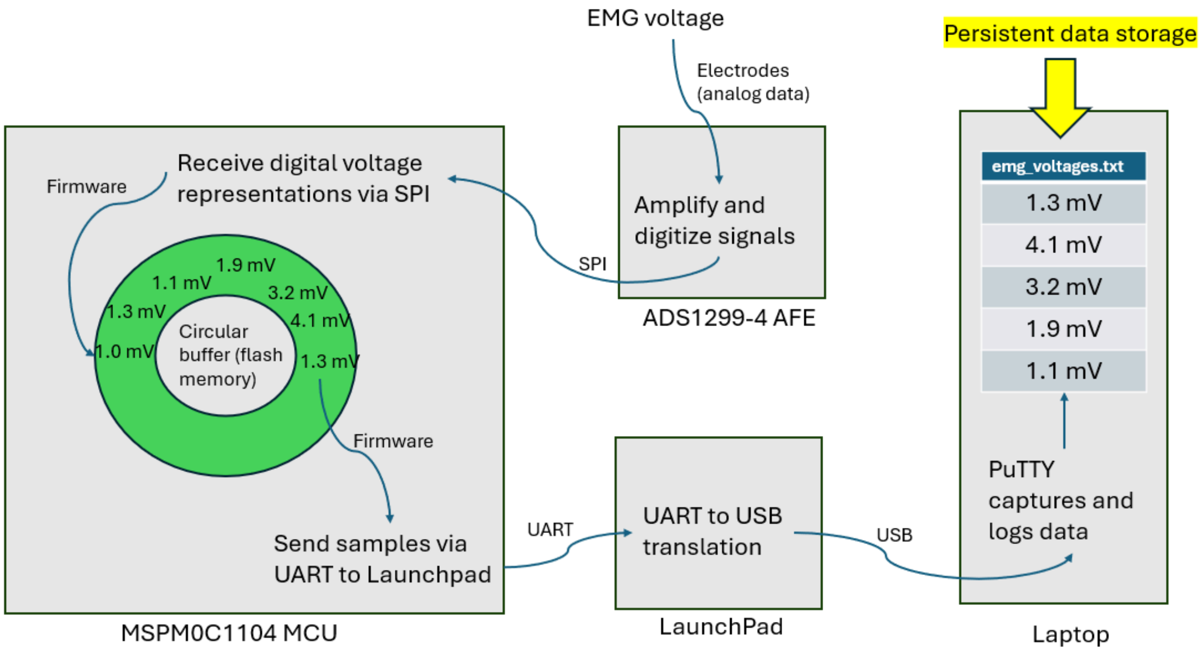


Figure 25: Connections between the firmware developer's laptop, the Launchpad and the DTCP PCB enabling programming and UART communication between the devices.

The voltage samples will need to be stored in a circular buffer in device memory to not introduce sampling latency from the time needed to transmit data to the laptop through the LaunchPad. The SPI firmware interface will read the voltage samples from the ADS1299 AFE and place them in the buffer (advancing the write pointer) while the UART interface will be printing the samples to PuTTY at the same time (advancing the read pointer).

c. Internal Systems

The following flowchart shows how the data is handled by the internal systems.

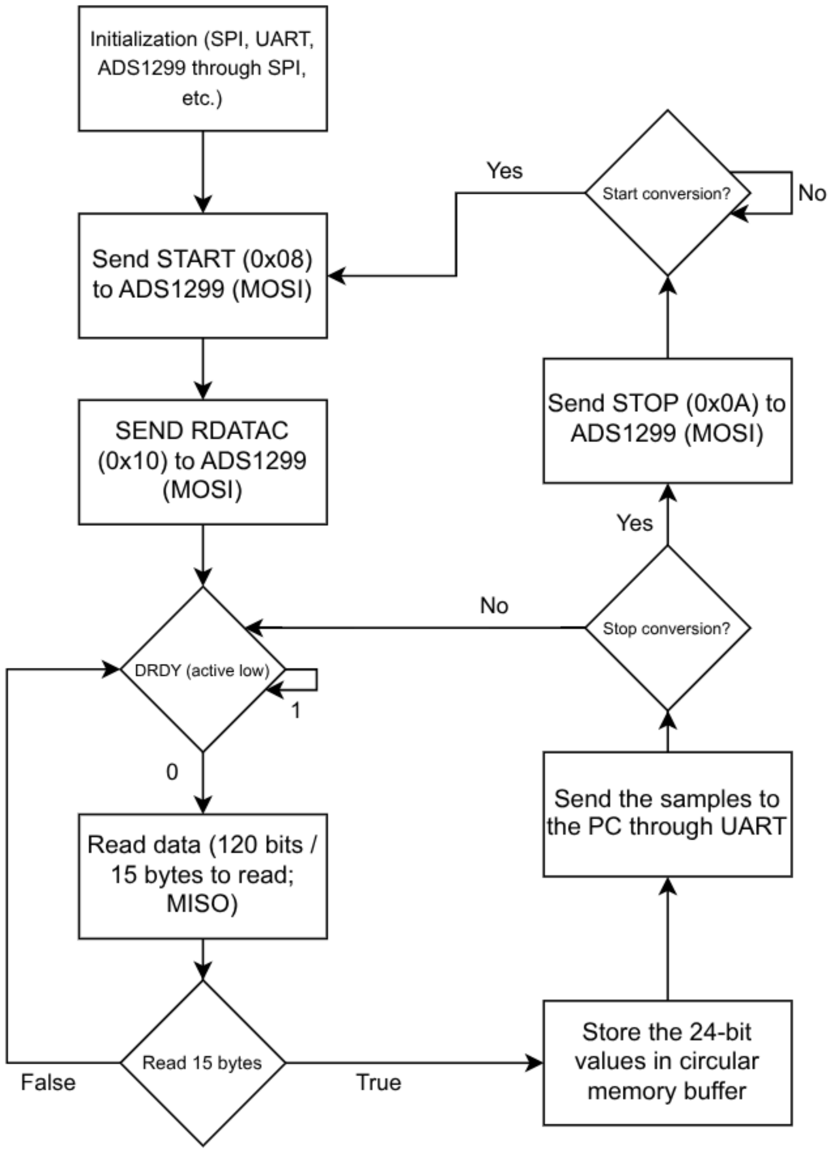


Figure 26: Data handling flowchart

3. Information Handling

a. Communication

As shown in figures 23-26, the ADS1299 microcontroller will use SPI to communicate with the AFE, and JTAG and UART to communicate with the LaunchPad. The LaunchPad will use USB serial data to communicate with Code Composer Studio and PuTTY on the user's laptop.

b. Integrity & Resilience

Because the DTCP implant power transmitter is a single-user research device, data corruption or malicious intrusion are not concerns. However, as a part of signal processing, the microprocessor will implement basic sanity checking on the received EMG data to ensure the received signals actually represent muscle activity rather than noise. Removing noise from the system is an important part of the design. The device should be isolated from 60 Hz power supply noise when it's battery powered, but it must be ensured that when data is being sent to a laptop, that device is also using a battery rather than being connected to a power supply that might propagate noise onto the PCB.

4. Project Repository and Video

GitHub repository: https://github.com/jdados/dtcp
Project overview video: https://youtu.be/saHZVm3RDYI

Bibliography

[1] M. Ludens, “RF Power Amplifier Design,” ludens.cl, 2020. https://ludens.cl/Electron/RFamps/RFamps.html

[2] Texas Instruments, “HF Power Amplifier (Reference Design Guide) RFID Systems / ASP 1.) Scope,” 2004. Accessed: Oct. 21, 2025. [Online]. Available: https://e2e.ti.com/cfs-file/__key/telligent-evolution-components-attachments/00-667-00-00-00-11-91-53/appnote_5F00_trf796x_5F00_pwramp_5F00_4w.pdf

[3] S. C. Cripps, RF Power Amplifiers for Wireless Communications, 2nd ed., Norwood, MA, USA: Artech House, 2006.

[4] F. H. Raab et al., “Power amplifiers and transmitters for RF and microwave,” IEEE Transactions on Microwave Theory and Techniques, vol. 50, no. 3, pp. 814–826, Mar. 2002. doi: 10.1109/22.989965

[5] N. O. Sokal and A. D. Sokal, “Class E-A new class of high-efficiency tuned single-ended switching power amplifiers,” *IEEE Journal of Solid-State Circuits*, vol. 10, no. 3, pp. 168–176, Jun. 1975, doi: https://doi.org/10.1109/jssc.1975.1050582.

[6] H. Dadashzadeh, “Design and implementation of class-E power amplifiers and rectifiers for a dual-band capacitive wireless power transfer system,” Doi.org, Oct. 2025, doi: https://doi.org/10.14288/1.0406624.

[7] Bernabe Rodríguez-Tapia et al. “Myoelectric Interfaces and Related Applications: Current State of EMG Signal Processing–A Systematic Review”. In: IEEE Access 8 (2020), pp. 7792–7805. DOI: 10.1109/ACCESS.2019.2963881.